

Exam Date & Time: 03-May-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL / MAY 2024

SUBJECT: ELE 1071/ELE_1071 - BASIC ELECTRICAL TECHNOLOGY

BASIC ELECTRICAL TECHNOLOGY [ELE 1071-CHM]

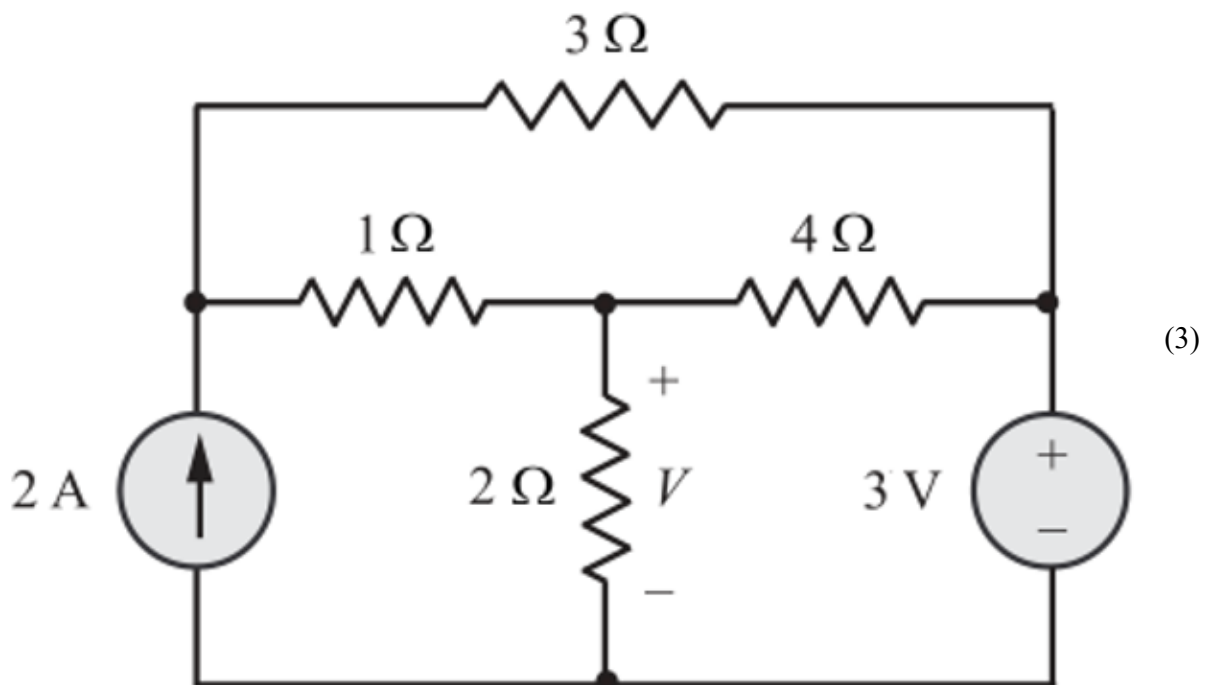
Marks: 50

Duration: 180 mins.

A

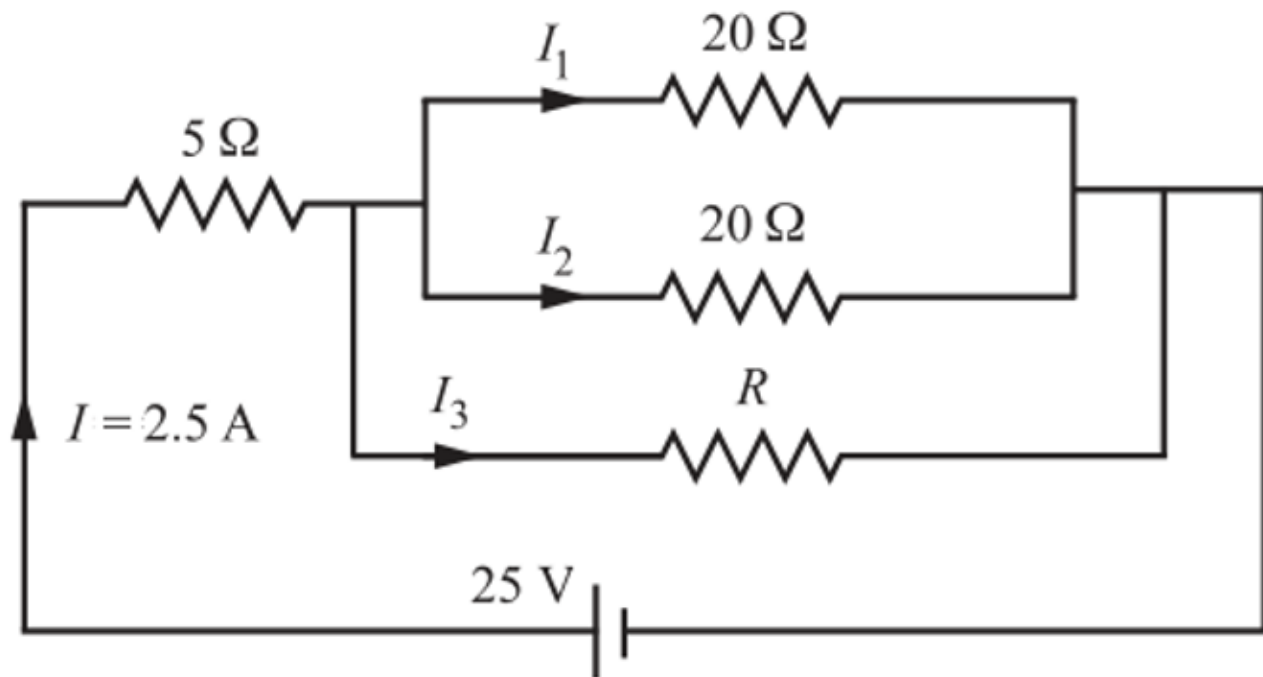
Answer all the questions.

1A) Determine the voltage V across the $2\ \Omega$ resistor in the given network.

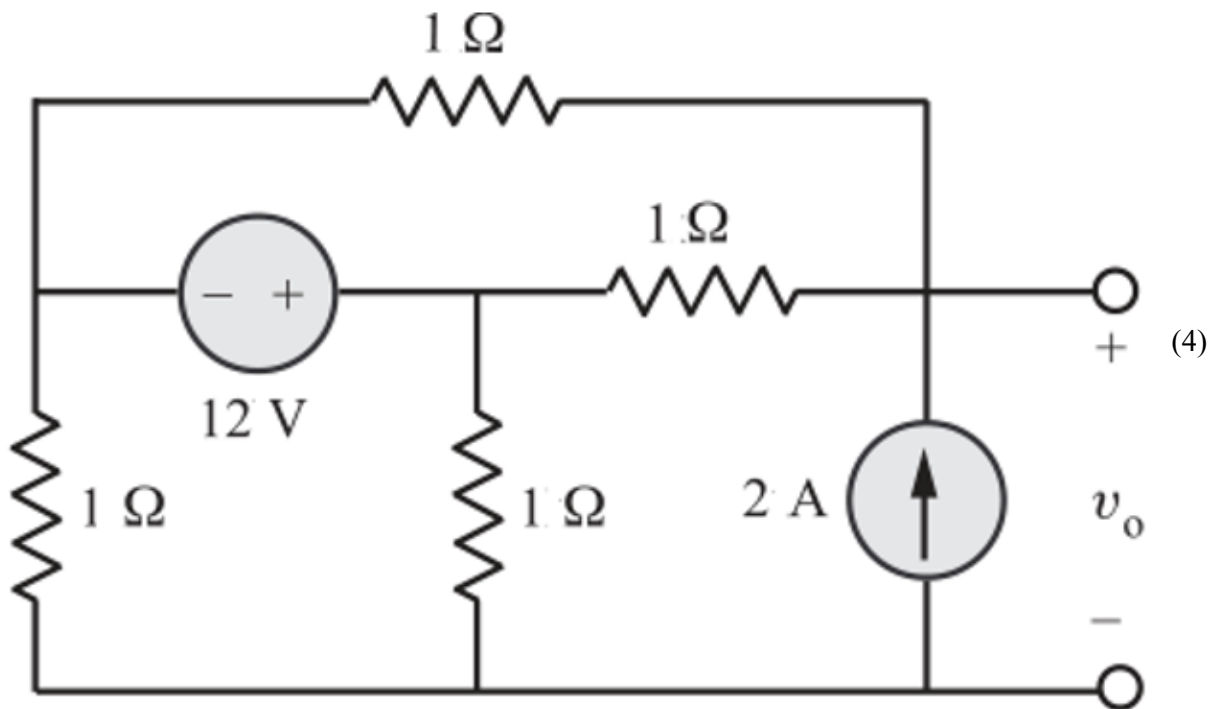


1B) Determine the value of R in the electrical circuit shown.

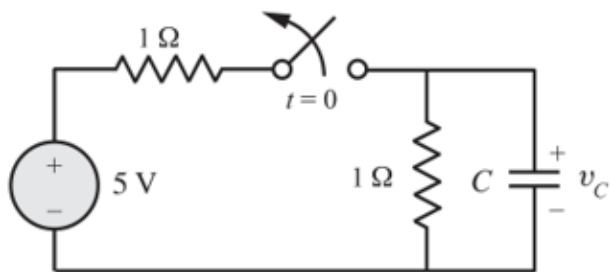
(3)



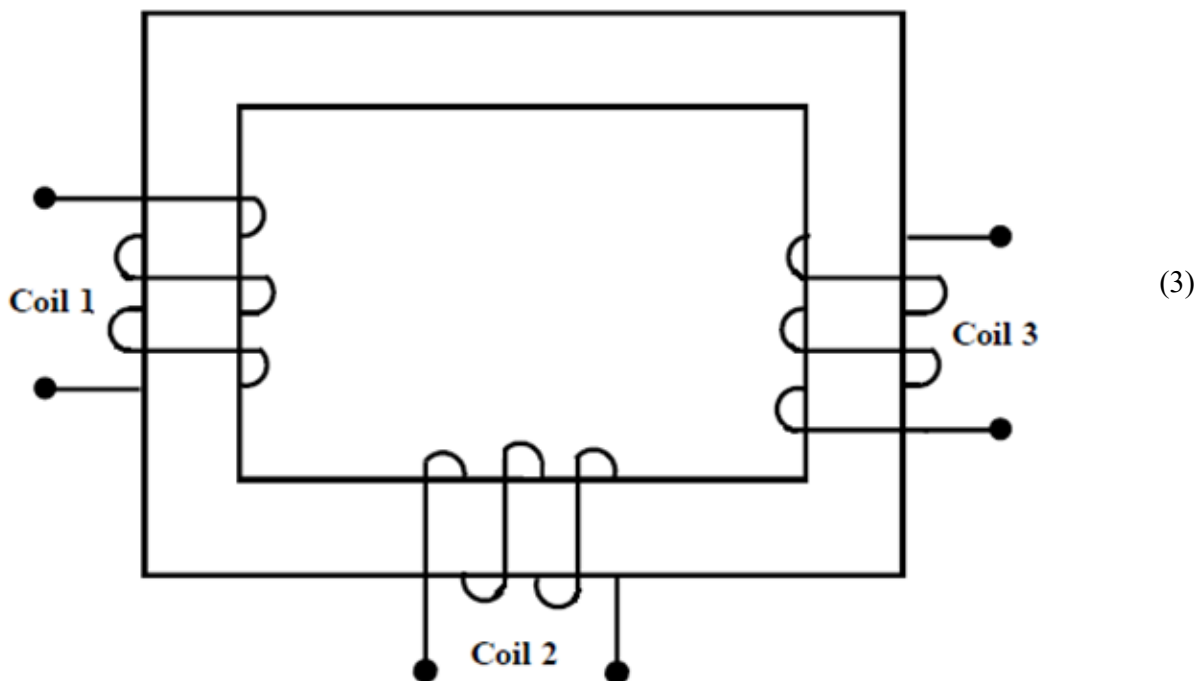
- 1C) Using **superposition** principle, determine the voltage v_0 as shown in the network.



- 2A) For the circuit shown, assume that the switch was in closed state for a long time. At $t = 0$, it is (3) operated as shown. Obtain $V_c(t)$ for $t > 0$. Also calculate the time when capacitor voltage becomes 1.25 V.



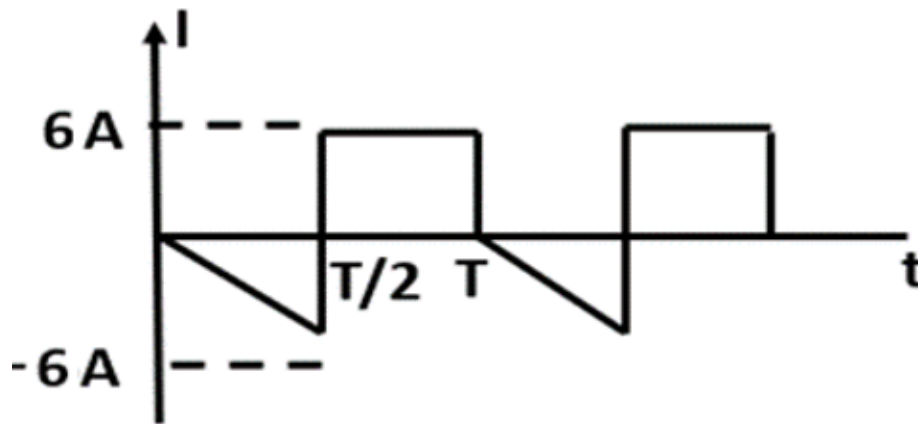
- 2B) Three identical coupled coils of self-inductance **40 mH** are wound on a rectangular core as shown. Calculate the maximum and the minimum equivalent inductances possible for this arrangement assuming that same current flows through all the coils. Given, the coupling coefficient, $K_{12} = 0.95$, $K_{23} = 0.9$, $K_{31} = 0.8$.



- 2C) A closed magnetic circuit of cast steel contains a **6 cm** long path of cross-sectional area **1 cm²** and a **2 cm** path of cross-sectional area **0.5 cm²**. A coil of **200** turns is wound around the **6 cm** length of the circuit and a current of **0.4 A** flows. Determine the flux density in the **2 cm** path, if the relative permeability of the cast steel is **750**. (4)

(Absolute permeability of air $= 4 \pi \times 10^{-7}$ H/m)

- 3A) Determine the **average** and **rms** values of the current waveform shown below. (3)



- 3B) A 1-phase circuit consists of a resistance of $10\ \Omega$, an inductance of $13\ \text{mH}$ and a capacitance of $150\ \mu\text{F}$ connected in series. A supply of $100\ \text{V}$ at $50\ \text{Hz}$ is given to the circuit. Calculate the impedance, power factor of the current and power consumed in the circuit. (3)
- 3C) Two branches are connected in parallel to a $230\ \text{V}$, $50\ \text{Hz}$, 1-phase AC supply.
Branch 1: A coil of resistance $85\ \Omega$ and inductance of $0.617\ \text{H}$
Branch 2: $35\ \Omega$ resistor in series with a $382\ \mu\text{F}$ capacitor
 Calculate,
 i) Circuit impedance (4)
 ii) Circuit admittance
 iii) Supply current
 iv) Net complex power and power factor
- 4A) A resistor and capacitor are in series with a variable inductor. When the circuit is connected to a $200\ \text{V}$, $50\ \text{Hz}$, 1-phase supply, the maximum rms current obtainable by varying the inductor is $314\ \text{mA}$. The voltage across the capacitor is then $300\ \text{V}$. Determine the circuit constants R , L and C . (4)
- 4B) A **star** connected load of $(5 + j\ 10)\ \Omega$ per phase is connected to a balanced 3-phase, $380\ \text{V}$, **ABC** supply. Assuming V_{AB} as reference phasor obtain the line currents, load voltages, active power, reactive power, and total volt ampere. (6)
- 5A) **Two** wattmeters are connected to measure the power input to a **3-phase** circuit, indicating **$2.5\ \text{kW}$** and **$0.5\ \text{kW}$** , respectively. Find the power factor of the circuit when,
 i) Both readings are positive (4)
 ii) Readings obtained after reversing the connection to the current coil of one instrument.
- 5B) Sketch a one-line diagram of a power system network architecture indicating voltage levels at various stages. (3)
- 5C) Explain the operation of a transformer. Obtain the expressions for primary and secondary induced EMF in a transformer. (3)

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